Lesson 13D Planets and Satellites: Kepler's Laws

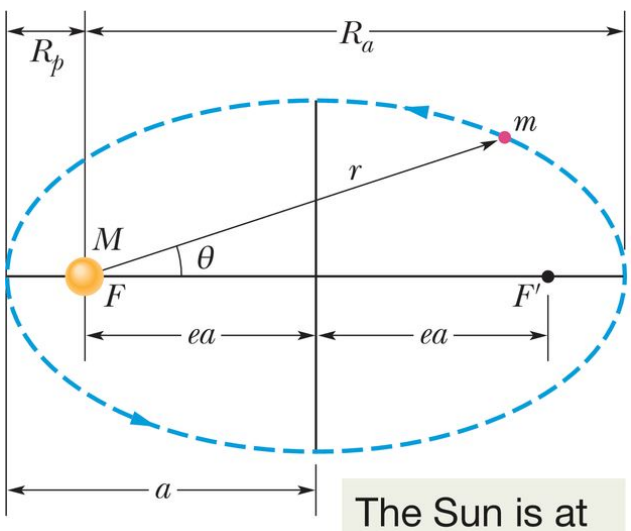
Kepler's Laws of Planetary Motion

The motion of planets in the solar system puzzled astronomers for centuries. Complex observations—like planets appearing to loop backward in the sky—required explanation beyond simple circular motion.

Johannes Kepler (1571–1630) used the precise data collected by **Tycho Brahe** (1546–1601) to derive three empirical laws that describe planetary motion:

Kepler's First Law – Law of Ellipses

Planets move in **elliptical orbits** with the Sun at one focus of the ellipse.



The Sun is at one of the two focal points.

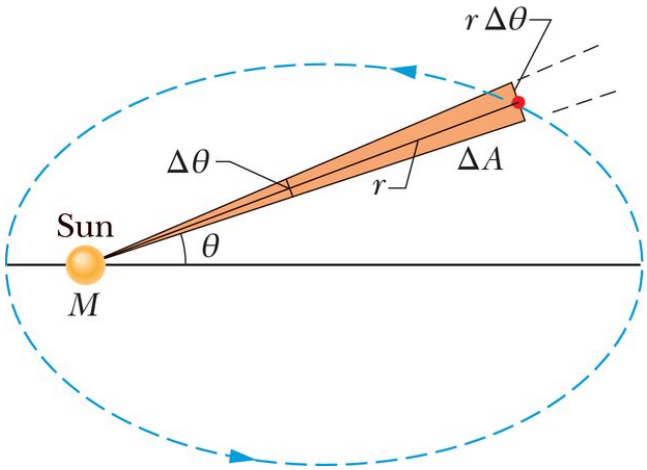
- Each orbit is described by:
  - **Semimajor axis** a: half the longest dimension of the ellipse
  - **Eccentricity** e: a measure of how elongated the ellipse is
    - $e = 0$ is a perfect circle
    - Earth's orbit has a small eccentricity: $e = 0.0167$

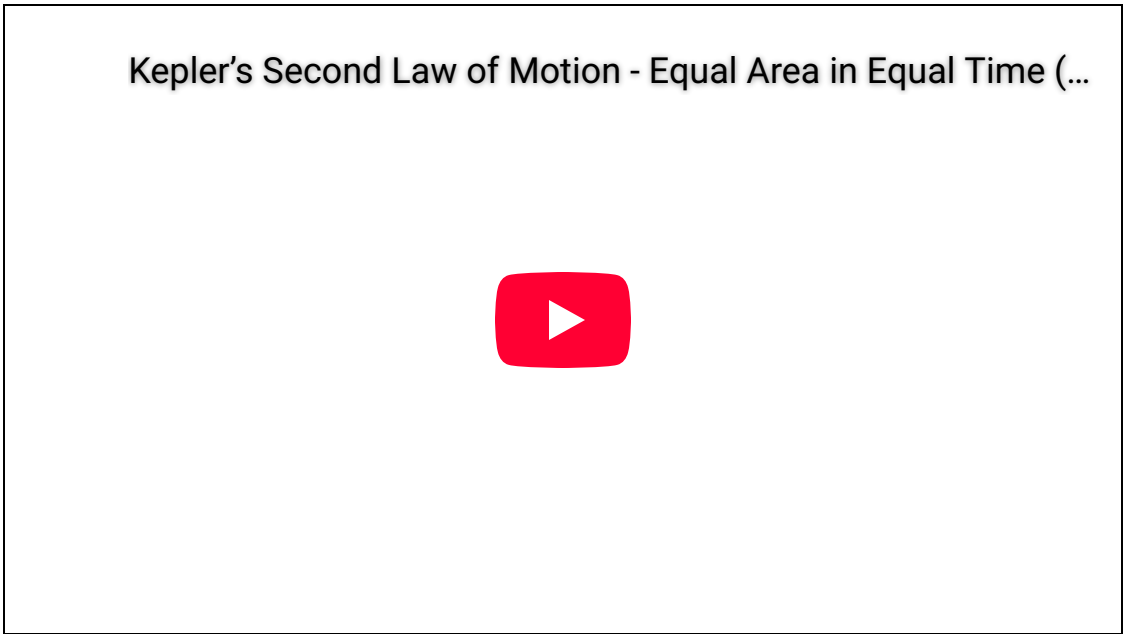
Kepler's First Law of Motion - Elliptical Orbits (Astronomy)

Kepler's Second Law – Law of Equal Areas

A line connecting a planet to the Sun sweeps out **equal areas in equal times**.

- This means that a planet moves **faster when closer to the Sun** (perihelion) and **slower when farther** (aphelion).
- This law reflects conservation of angular momentum: as the distance to the Sun decreases, the orbital speed increases.





Kepler's Third Law – Law of Periods

The square of the orbital period T of a planet is proportional to the cube of the semimajor axis a:

$$T^2 \propto a^3$$

- This relation holds for any object orbiting under gravity, including satellites.
- For circular orbits, we may use the orbital radius r; for ellipses, use the semimajor axis a.

The equation for **Kepler's Third Law (Law of Periods)** is:

$$T^2 = \frac{4\pi^2}{GM} a^3$$

Where:

- T is the orbital period
- a is the semimajor axis of the orbit
- G is the gravitational constant
- M is the mass of the central body (e.g., the Sun)

Satellites: Orbits and Energy

We can also describe orbits using energy and Newton's law of gravitation. For a satellite in a circular orbit:

- The gravitational force provides the centripetal acceleration.
- By equating gravitational and centripetal forces:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

- Solving for kinetic energy:

$$K = \frac{1}{2}mv^2 = \frac{GMm}{2r}$$

- Gravitational potential energy is:

$$U = -\frac{GMm}{r}$$

- Total mechanical energy:

$$E = K + U = -\frac{GMm}{2r}$$

This shows that:

- The total energy is negative, indicating a bound orbit
- The total energy depends only on the orbital radius (or semimajor axis a), not on eccentricity

For elliptical orbits, we generalize:

$$E = -\frac{GMm}{2a}$$

This means:

- All orbits with the same semimajor axis a have the same total energy, regardless of shape.
- This is true for both planets and satellites.